

Raghav Chaturvedi

Julius-Maximilians-Universität, Sanderring 2, 97070 Würzburg, Germany

+49 1522 8265232 ✉ raghav.chaturvedi@uni-wuerzburg.de [linkedin.com/raghav-chaturvedi97](https://www.linkedin.com/raghav-chaturvedi97)

🎓 [Google Scholar Profile](#) **Date of Birth:** 25th September 1997 **Place of Birth:** Delhi, India **Nationality:** Indian

Current Position

Julius-Maximilians-University (JMU)

Würzburg, Germany

Postdoctoral Researcher / Research Associate (Theoretical Condensed Matter Physics)

Oct 2024–Present

- Advisor: Prof. Dr. Ewelina M. Hankiewicz
- **Current Research Areas:**
 - * Quantum transport in topological, spin-dependent, and disordered condensed-matter systems.
 - * Non-Hermitian topology, real-space topological diagnostics, finite-size effects, and experimental collaborations.
 - * Machine-learning-assisted physics: finite-size convergence of real-space topological invariants.

Education

Leibniz Institute for Solid State and Materials Research (IFW)

Dresden, Germany

Doctor of Philosophy (Theoretical Condensed Matter Physics)

Oct 2020–February 2025

- Thesis title: Non-Hermitian topology of transport in quantum Hall systems. Grade: Magna Cum Laude
- Advisors: Prof. Dr. Jeroen van den Brink (IFW), Prof. Dr. Ewelina M. Hankiewicz (JMU), and Dr. Ion Cosma Fulga (IFW)
- Coursework (20 credits): Many body theory in condensed matter physics, Geometry and topology in quantum physics, Analytical tools for quantum many-body physics, Magnetism at nanoscale, Practical quantum communication networks, Machine learning and data science, Statistical evidence in experimental science
- Teaching Assistant: 2 Semesters for Grundpraktikum III

Indian Institute of Science Education and Research

Aug 2015–Aug 2020

Bachelor of Science–Master of Science Dual Degree (CGPA 8.77/10)

Trivandrum, India

- Thesis title: Investigating metal-insulator transitions in twisted bilayer graphene using Z2 Slave Spin theory; Advisor: Prof. Dr. Amal Medhi.
- 185 Credits, Physics Major, Mathematics Minor, 3rd in a class of 140

Publications and Preprints

- Machine learning prediction of the convergence criterion for a topological invariant of finite non-Hermitian chains – [Raghav Chaturvedi](#), Viktor Könye, Ewelina M. Hankiewicz. [arXiv:2607.05900 \(2026\)](#). [\[Link\]](#)
- Non-Hermitian topology of quantum spin-Hall systems to detect edge-state polarization – [Raghav Chaturvedi](#), Ion Cosma Fulga, Jeroen van den Brink, Ewelina M. Hankiewicz. [arXiv:2602.12048 \(2026\)](#). [\[Link\]](#)
- Non-Hermitian topology of transport in the quantum Hall phases in graphene – [Raghav Chaturvedi](#), Viktor Könye, Ewelina M. Hankiewicz, et al. **Phys. Rev. B** **111**, 245424. [\[Link\]](#)
- Non-Hermitian topology in a multi-terminal quantum Hall device – Kyrylo Ochkan*, [Raghav Chaturvedi*](#), Viktor Könye, et al. **Nat. Phys.** **20**, 395–401 (2024). * Joint first author. [\[Link\]](#)
- Non-Hermitian topology in the quantum Hall effect of graphene – Burak Özer*, Kyrylo Ochkan*, [Raghav Chaturvedi](#), et al. [arXiv:2410.14329 \(under review in Nat. Commun.\)](#). Second author. [\[Link\]](#)
- Spin dynamics investigations of multifunctional ambient scalable Fe₃O₄ surface decorated ZnO magnetic nanocomposite using FMR – Saurabh Pathak, Rajni Verma, Sakshi Singhal, [Raghav Chaturvedi](#), Prashant Kumar, Pragati Sharma, Rajendra Prasad Pant, Xu Wang. **Scientific Reports** **11**, 3799 (2021). [\[Link\]](#)
- Measurement of Static and Dynamic Magneto-Viscoelasticity in Facile Varying pH Synthesized CoFe₂O₄-Based Magnetic Fluid – Akash Mishra, Saurabh Pathak, Prashant Kumar, Arjun Singh, Komal Jain, [Raghav Chaturvedi](#), Dinesh Singh, GA Basheed, RP Pant. **IEEE Transactions on Magnetics** (2019). [\[Link\]](#)

Manuscripts under preparation

- Spin Hall response and nonlocal resistance sign changes in HgTe devices – [Raghav Chaturvedi](#), Shyam Sundar Yadav, Hartmut Buhmann, Lorenz W. Molenkamp, Ewelina M. Hankiewicz.
- Critical exponent for the Hatano-Nelson chain using a real-space topological invariant – [Raghav Chaturvedi](#), Ion Cosma Fulga.
- Localizer index for non-Hermitian topological systems based on pseudo-spectra methods – [Raghav Chaturvedi](#), Ion Cosma Fulga.

Certifications

- Machine Learning Specialization (Stanford University – Coursera)
Covered supervised/unsupervised learning, regression, classification, clustering, recommender systems; applied with Python, NumPy, and scikit-learn. [\[Link\]](#)
- Deep Learning Specialization (DeepLearning.AI – Coursera)
Implemented deep neural networks, CNNs, RNNs, optimization methods, and regularization in TensorFlow and PyTorch. [\[Link\]](#)
- Introduction to High-Performance and Parallel Computing (Univ. of Colorado Boulder – Coursera)
Learned MPI, OpenMP, GPU basics, and performance scaling concepts; practiced HPC job submission with Slurm. [\[Link\]](#)

Academic Accolades

- KVPY Scholar: Fellowship awarded by Indian Institute of Science, Bangalore, India to top 2000 students in India.
- DST-INSPIRE Scholar: Fellowship awarded by Govt. of India to the top 1 percent of Indian students.

Conferences and Schools

Contributed Talks |

- DPG meeting Dresden, Germany, March 2026: 12 minute talk about theoretical basis of Non-Hermitian topology in graphene.
- APS March Meeting Minneapolis, USA, March 2024: 10 minute talk about Non-Hermitian topology in a multi-terminal quantum Hall device.
- IFW Theory Meeting August 2023: 50 minute talk about theoretical basis of Non-Hermitian topology in Chern insulators.
- DPG meeting Regensburg, Germany, September 2022: 12 minute talk about theoretical basis of Non-Hermitian topology in a multi-terminal quantum Hall setup.

Poster Presentations |

- CT.QMAT24 – International Conference on Complexity and Topology in Quantum Matter Dresden, Germany, September 2024
- QMA retreat meeting Weimar 2024
- IFW winter school, Potsdam, Germany, February 2023
- CT.QMAT22 – International Conference on Complexity and Topology in Quantum Matter Würzburg, Germany, July 2022
- First Fall School “Topological Quantum Matter”, Erfurt, Germany, October 2021

Past research experience (Masters)

- **Summer Research Fellow (May 2019–July 2019)** Title: Investigating the electronic properties of twisted multi-layer graphene systems. Advisor: Dr. Igor Gornyi (Mirlin Group), Karlsruhe Institute of Technology, Karlsruhe, Germany
- **Indian Academy of Sciences Summer Research Fellow (May 2018–July 2018)** Title: Review of topological phases of matter and non-abelian anyon statistics. Advisor: Prof. Dr. Sumathi Rao, Harish Chandra Institute Allahabad, India
- **Summer Research Fellow (May 2017–July 2017)** Title: Spin dynamics investigations in magnetic nanocomposites using FMR. Advisor: Dr. R.P. Pant, Chief Scientist, CSIR-National Physical Laboratory New Delhi, India
- **Summer Research Fellow (May 2016–July 2016)** Title: Magneto-viscoelasticity study of magnetic fluids. Advisor: Dr. R.P. Pant, Chief Scientist, CSIR-National Physical Laboratory New Delhi, India

Skills

- **Programming & Computing:** Python, NumPy, pandas, scikit-learn, PyTorch, TensorFlow, C++, Bash scripting, Git, Kwant, High-Performance Computing (MPI, Slurm), LaTeX.
- **Software:** Mathematica, Matlab, Origin, Inkscape, Adobe Premiere Pro.
- **Languages:** English (Proficient), German (B1).

References

Prof. Dr. Ewelina M. Hankiewicz – Associate Professor at Julius-Maximilians-University Würzburg. Email: ewelina.hankiewicz@physik.uni-wuerzburg.de, Phone +49 931 31-84998

Dr. Ion Cosma Fulga – Permanent Group Leader, Leibniz Institute for Solid State and Materials Research, IFW Dresden. Email: i.c.fulga@ifw-dresden.de, Phone +49-351-4659-737

Dr. Joseph Dufouleur – Group leader – Quantum Transport, Leibniz Institute for Solid State and Materials Research, IFW Dresden. E-mail: j.dufouleur@ifw-dresden.de